ML Mini Project: PvP Chat Sentiment Classification

Name: Arvind Ajay Bharadwaj **SRN:** PES1UG24AM052

Name: Amogh B Shetty **SRN:** PES1UG24AM034

Problem and Objective

Online game messages can express frustration without directly insulting another player. This project compares traditional machine learning and neural networks for classifying chat from DOTA 2 and League of Legends.

The dataset contains **33,000 messages**: 21,120 for training, 5,280 for validation, and 6,600 for testing. The three labels are **0: neutral/positive**, **1: negative attitude**, and **2: derogatory**. Each supplied split is balanced; the test set contains 2,200 messages per class.

Accuracy measures the proportion of correct predictions. Macro F1 averages the F1 scores of all classes equally, balancing precision and recall.

Implementation

Traditional models include SVM, Naive Bayes, K-nearest neighbours (KNN), and Random Forest. These use TF-IDF features, except for a Naive Bayes baseline that uses a fixed word list. SVM performs binary classification by combining negative-attitude and derogatory messages into one negative class.

The neural experiments use Word2Vec and GloVe embeddings with 50 and 100 dimensions. The feedforward network averages word embeddings before applying dense layers. The LSTM processes the words sequentially.

Four embeddings and three configurations per model produce 24 neural experiments. Training uses frozen embeddings, Adam with a learning rate of 0.001, batch size 128, and dropout 0.2. Neural checkpoints are selected using validation macro F1, with early stopping after 25 epochs without improvement.

Results

Accuracy measures the proportion of correct predictions. Macro F1 gives equal importance to each class by averaging their F1 scores. All values below are percentages.

Three-class model	Accuracy	Macro F1
Naive Bayes: word-list baseline	92.38	92.34
Naive Bayes: TF-IDF	91.77	91.74
KNN: K = 6	79.44	79.17
Random Forest: Gini	94.39	94.35
Random Forest: entropy	94.23	94.19
Random Forest: entropy, depth 32	91.24	91.13
Feedforward: Word2Vec 100d, ReLU + ReLU	93.15	93.15
LSTM: Word2Vec 100d, 128 units, softmax	96.61	96.61

The neural rows show the validation-selected models. K = 6 is the highest-scoring reported KNN setting among K = 1, 2, 3, and 6.

The LSTM achieves the highest reported three-class macro F1. It exceeds Random Forest (Gini) by 2.26 percentage points and the selected feedforward network by 3.46 percentage points.

For the separate binary task, SVM achieves accuracies of 98.85% with a linear kernel, 98.67% with an RBF kernel, and 95.64% with a polynomial kernel. These scores should not be compared directly with the three-class results.

Embedding comparison

The following comparison holds the network configuration fixed. Values are test macro F1 percentages.

Configuration	W2V 50d	W2V 100d	GloVe 50d	GloVe 100d
Feedforward: ReLU + ReLU	89.82	93.15	86.50	91.08
LSTM: 64 units, softmax	95.79	96.21	95.89	96.45

W2V denotes Word2Vec. Across all matched configurations, 100-dimensional embeddings outperform 50-dimensional embeddings. Word2Vec performs better for the two-layer ReLU feedforward network, while GloVe performs slightly better for the 64-unit softmax LSTM.

The selected feedforward checkpoint is from epoch 41, and the selected LSTM checkpoint is from epoch 136. Both use Word2Vec 100d.

Performance by class

The following table shows per-class F1 percentages.

Model	Class 0	Class 1	Class 2
Random Forest: Gini	96.09	92.07	94.91
Selected feedforward	95.02	90.84	93.58
Selected LSTM	97.62	95.43	96.77

Negative attitude is the hardest class for all three models shown. It can overlap with ordinary frustration and direct insults. The LSTM improves F1 for every class compared with the selected feedforward network, suggesting that word order and context are useful.

Limitations

This is a reimplementation rather than an exact numerical reproduction. The original paper reports an uneven class distribution, while the supplied splits are balanced.

The paper's preparation of 50d and 100d Google News Word2Vec embeddings is underspecified. This implementation reduces the public 300d vectors using PCA fitted on the training vocabulary.

The neural experiments use one random seed, so small differences should not be treated as statistically significant. Four normalized messages also overlap between training and test. Further evaluation should use multiple seeds and prevent duplicate messages from crossing splits.

Conclusion

The validation-selected LSTM is the strongest reported three-class model, achieving 96.61% accuracy and macro F1. Random Forest (Gini) is a competitive traditional alternative at 94.35% macro F1.

The experiments favour 100-dimensional embeddings, while negative-attitude messages remain the most difficult to classify. The saved checkpoints also support a live demonstration that classifies typed messages without retraining.